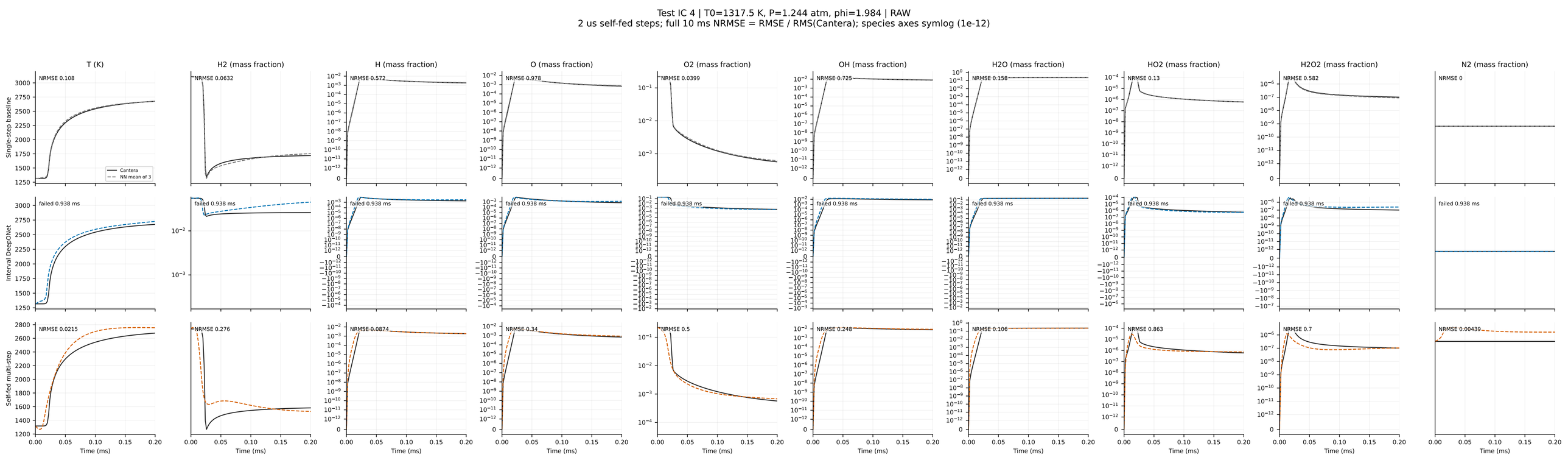
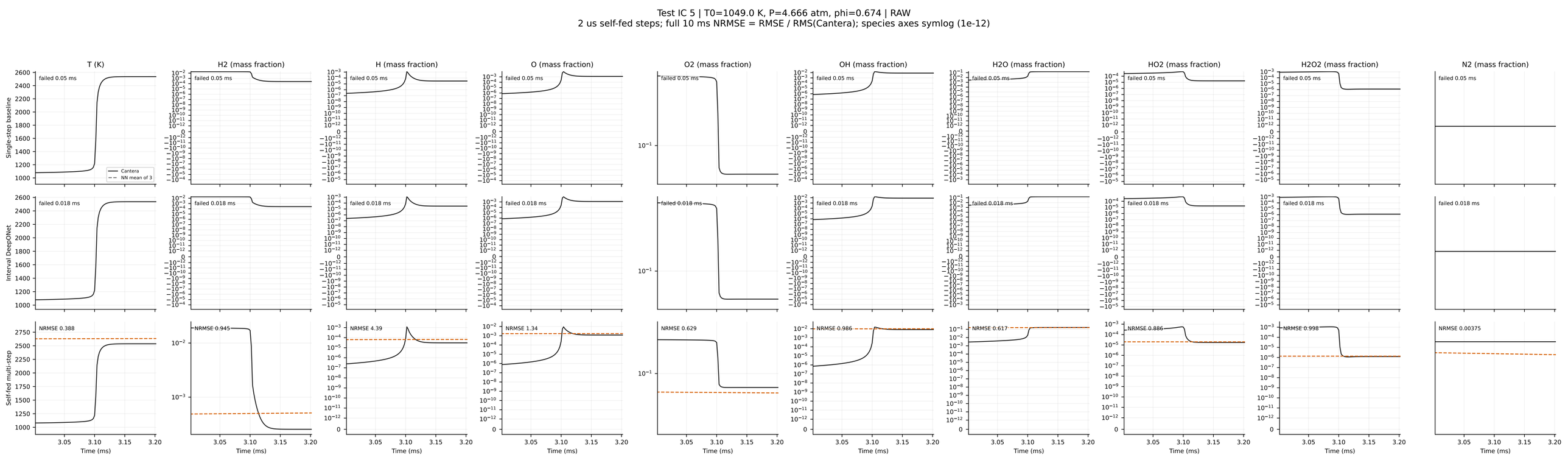
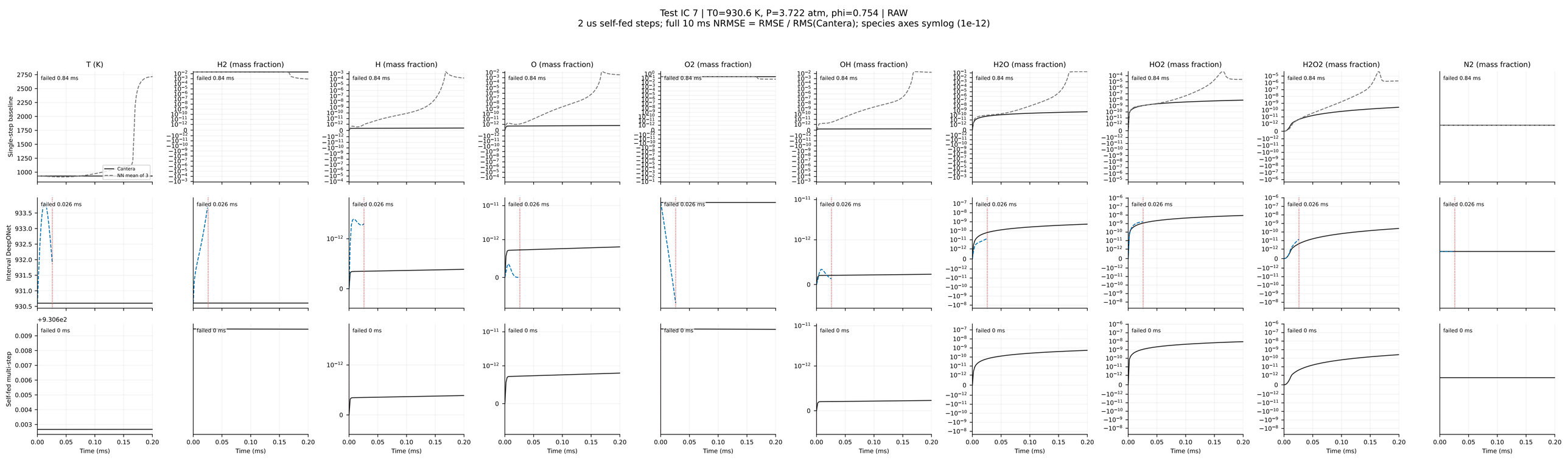
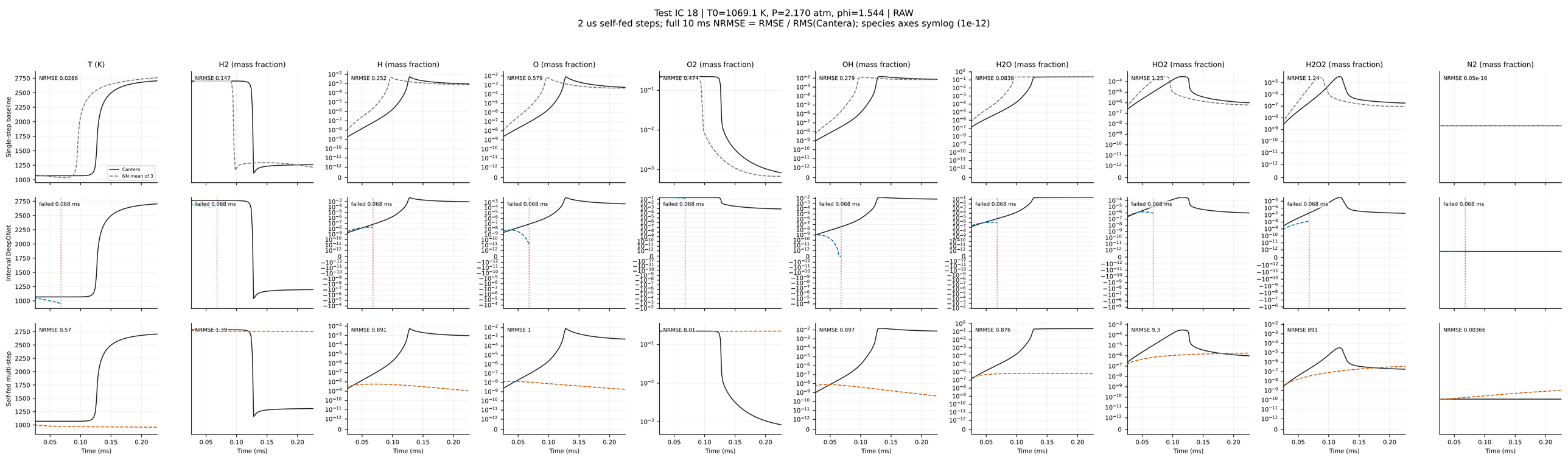
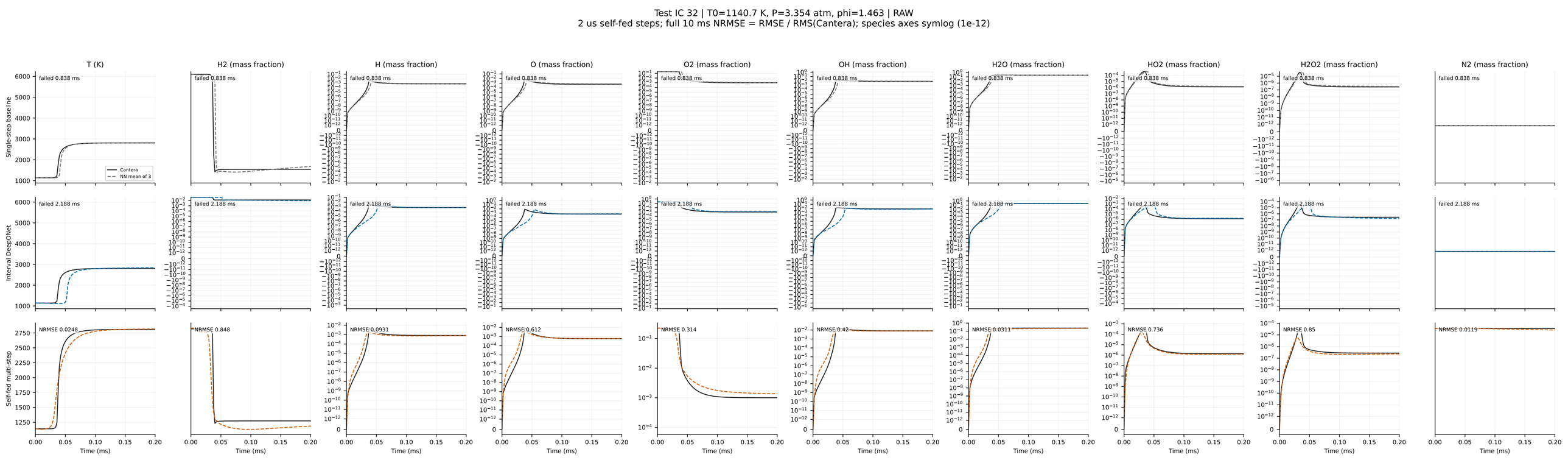
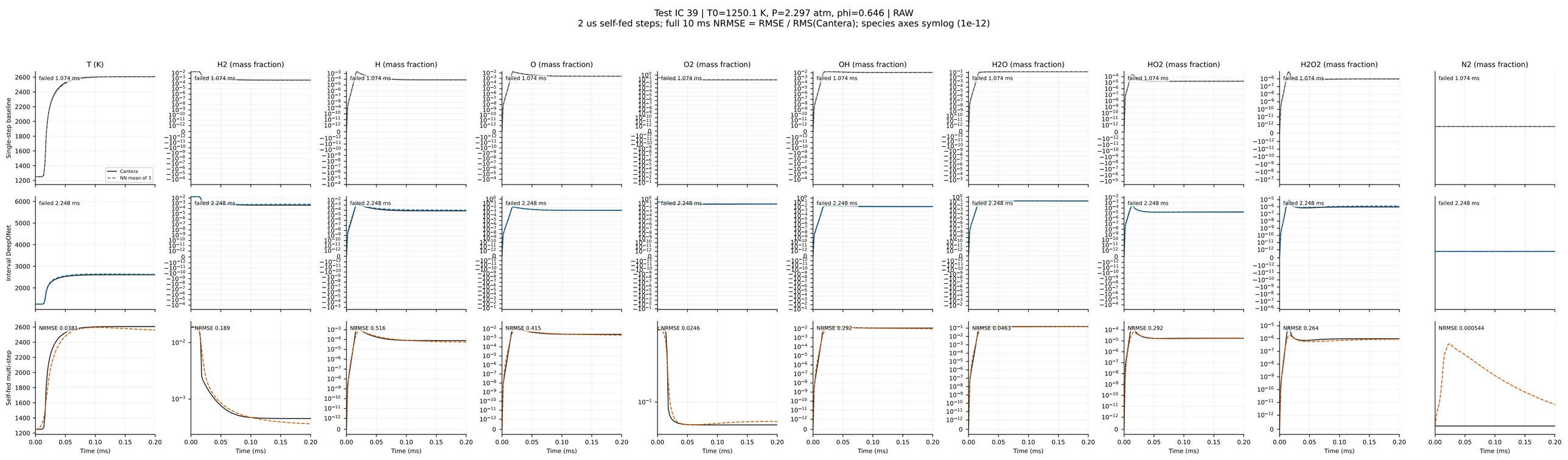

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.108
NRMSE 0.0632
NRMSE 0.572
NRMSE 0.978
NRMSE 0.0399
NRMSE 0.725
NRMSE 0.158
NRMSE 0.13
NRMSE 0.582
NRMSE 0
Cantera
NN mean of 3
failed 0.938 ms
NRMSE 0.0215
NRMSE 0.276
NRMSE 0.0874
NRMSE 0.34
NRMSE 0.5
NRMSE 0.248
NRMSE 0.106
NRMSE 0.863
NRMSE 0.7
NRMSE 0.00439
Time (ms)

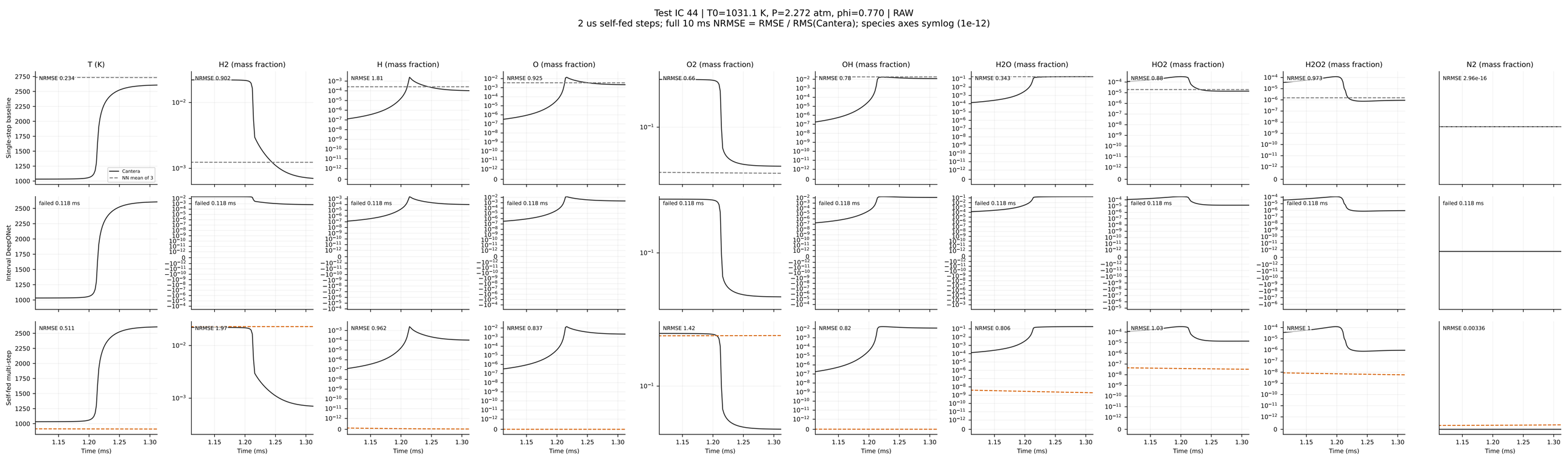

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.05 ms
failed 0.018 ms
Cantera
NN mean of 3
NRMSE 0.388
NRMSE 0.945
NRMSE 4.39
NRMSE 1.34
NRMSE 0.629
NRMSE 0.986
NRMSE 0.617
NRMSE 0.886
NRMSE 0.998
NRMSE 0.00375
Time (ms)

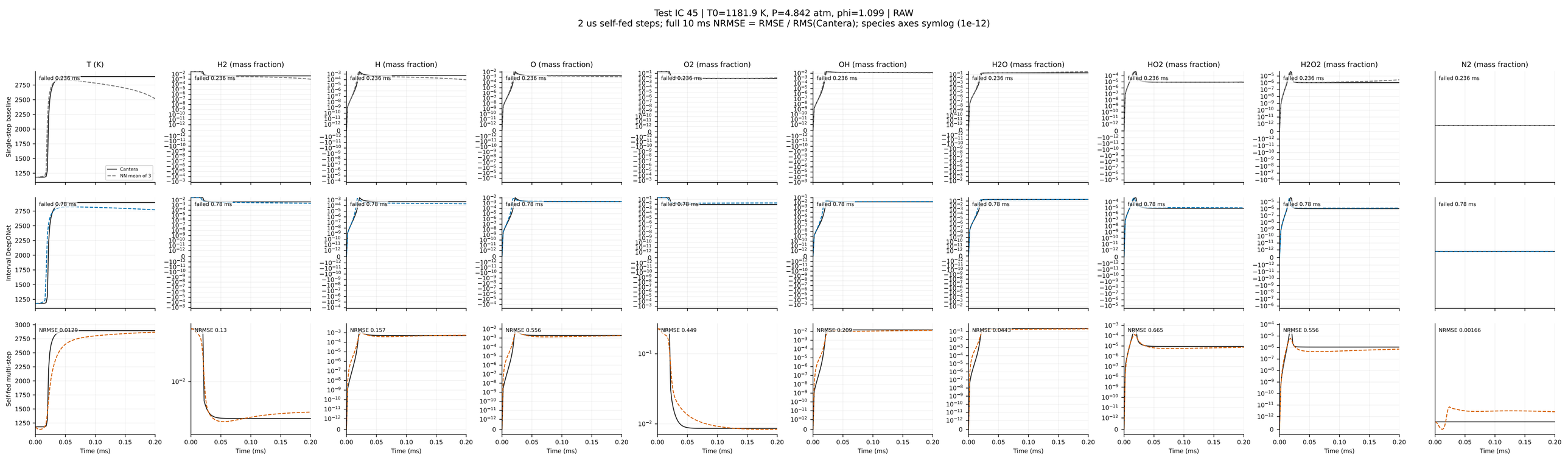

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.84 ms
failed 0.026 ms
failed 0 ms
Cantera
NN mean of 3
+9.306e2
Time (ms)

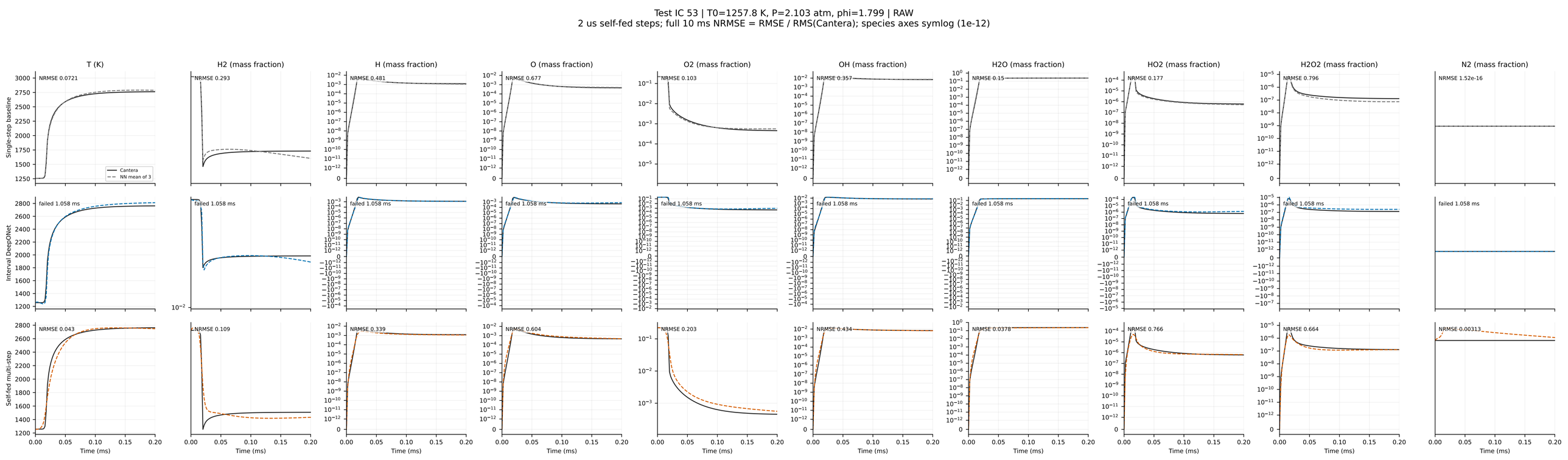

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0286
NRMSE 0.147
NRMSE 0.252
NRMSE 0.579
NRMSE 0.474
NRMSE 0.279
NRMSE 0.0836
NRMSE 1.25
NRMSE 1.24
NRMSE 6.05e-16
failed 0.068 ms
NRMSE 0.57
NRMSE 1.39
NRMSE 0.891
NRMSE 1
NRMSE 8.01
NRMSE 0.897
NRMSE 0.876
NRMSE 9.3
NRMSE 891
NRMSE 0.00366
Time (ms)

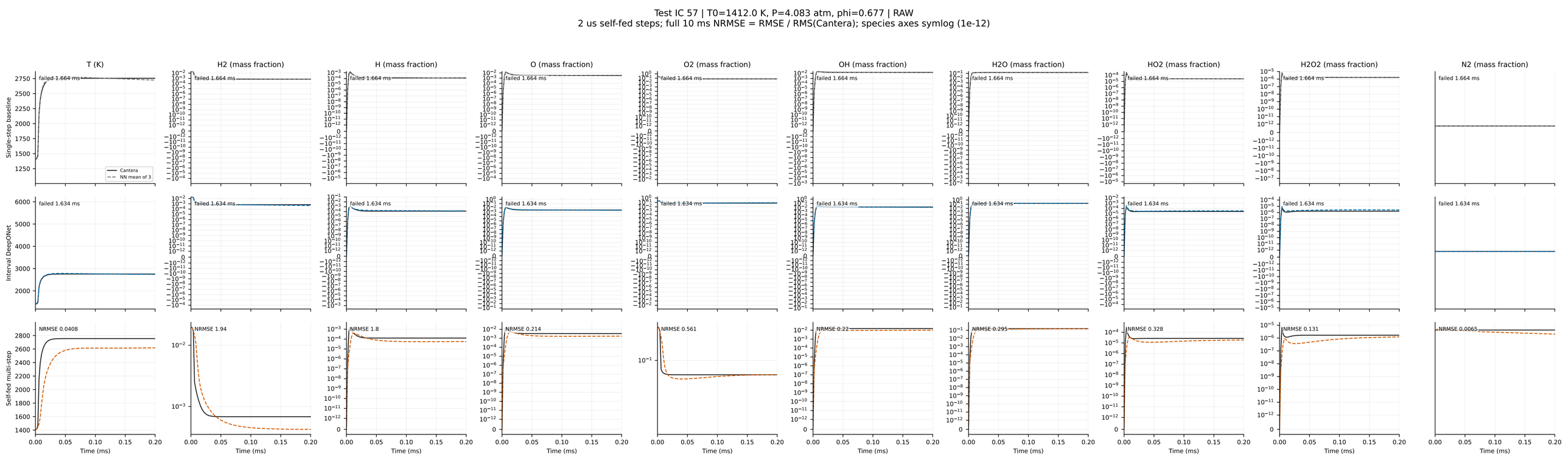

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.838 ms
failed 2.188 ms
Cantera
NN mean of 3
NRMSE 0.0248
NRMSE 0.848
NRMSE 0.0931
NRMSE 0.612
NRMSE 0.314
NRMSE 0.42
NRMSE 0.0311
NRMSE 0.736
NRMSE 0.85
NRMSE 0.0119
Time (ms)

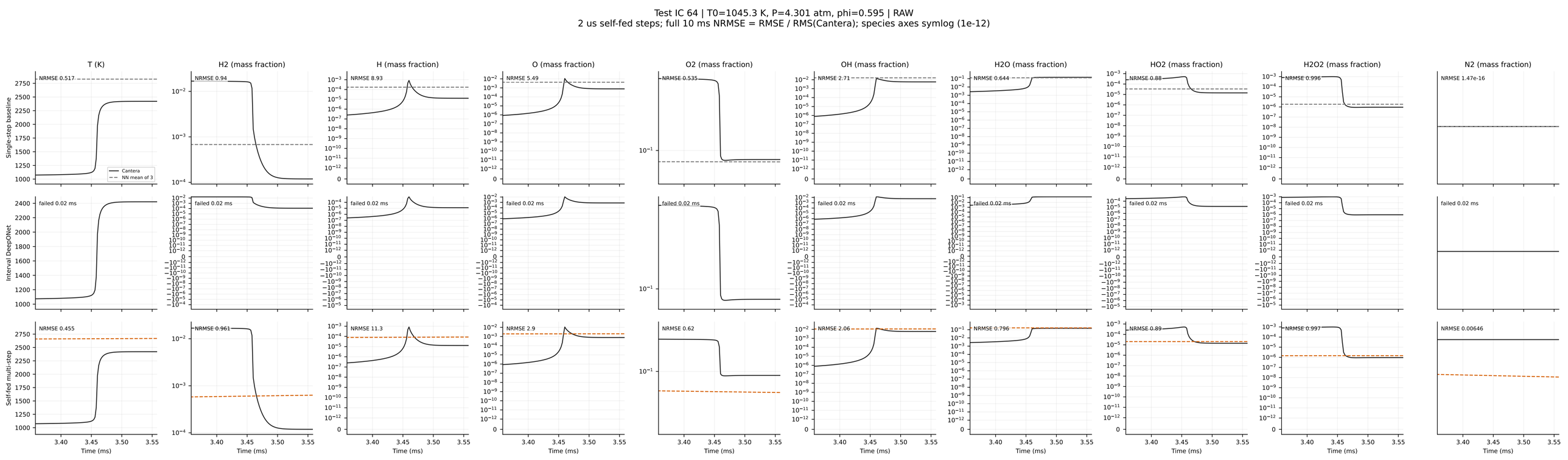

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 1.074 ms
failed 2.248 ms
Cantera
NN mean of 3
NRMSE 0.0381
NRMSE 0.189
NRMSE 0.516
NRMSE 0.415
NRMSE 0.0246
NRMSE 0.292
NRMSE 0.0463
NRMSE 0.292
NRMSE 0.264
NRMSE 0.000544
Time (ms)

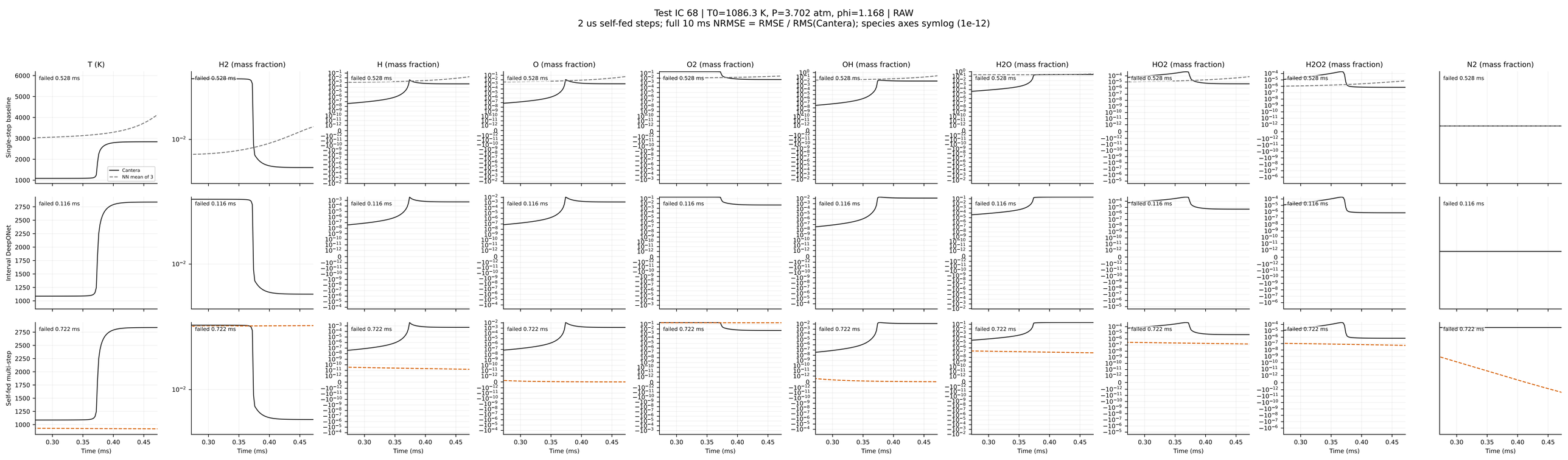

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.234
NRMSE 0.902
NRMSE 1.81
NRMSE 0.925
NRMSE 0.66
NRMSE 0.78
NRMSE 0.343
NRMSE 0.88
NRMSE 0.973
NRMSE 2.96e-16
Cantera
NN mean of 3
failed 0.118 ms
NRMSE 0.511
NRMSE 1.97
NRMSE 0.962
NRMSE 0.837
NRMSE 1.42
NRMSE 0.82
NRMSE 0.806
NRMSE 1.03
NRMSE 1
NRMSE 0.00336
Time (ms)

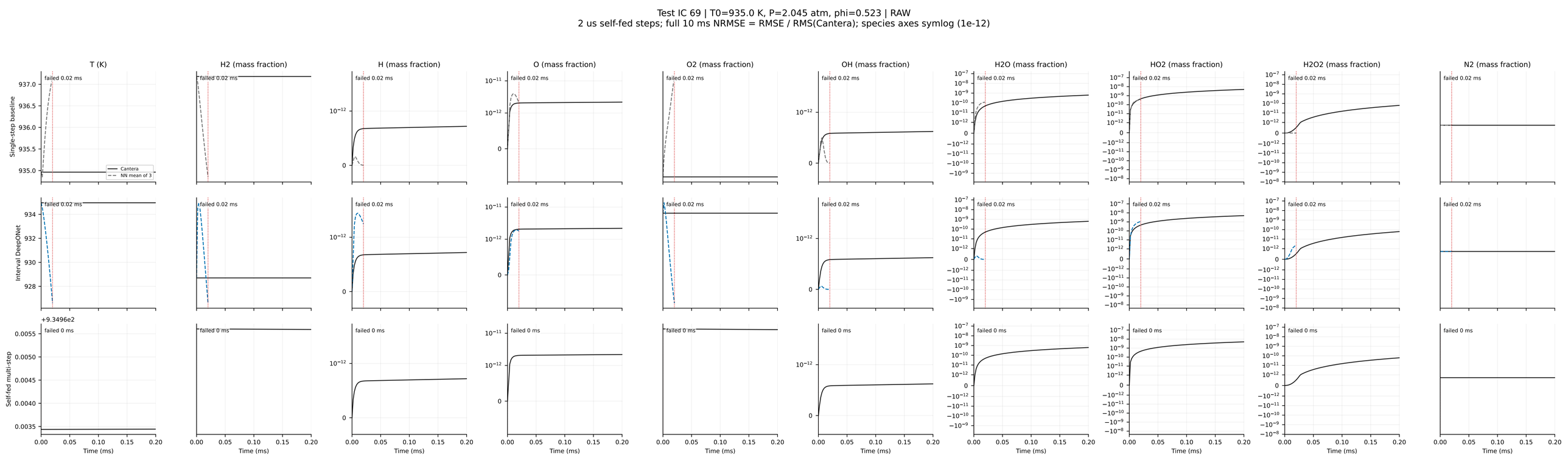

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.236 ms
failed 0.78 ms
Cantera
NN mean of 3
NRMSE 0.0129
NRMSE 0.13
NRMSE 0.157
NRMSE 0.556
NRMSE 0.449
NRMSE 0.209
NRMSE 0.0443
NRMSE 0.665
NRMSE 0.556
NRMSE 0.00166
Time (ms)

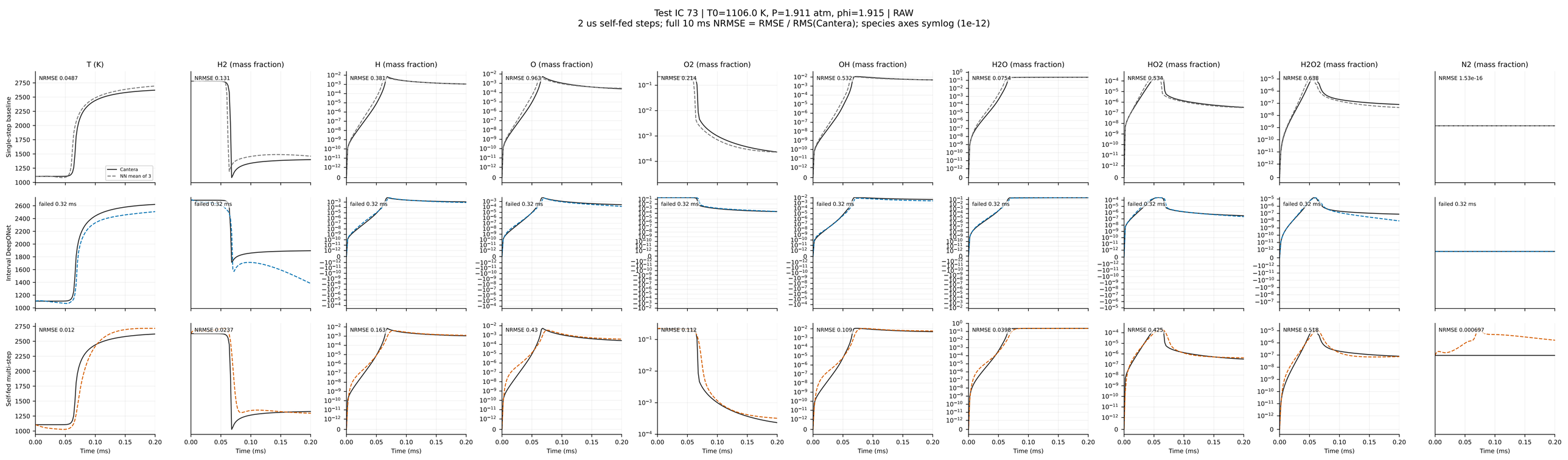

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0721
NRMSE 0.293
NRMSE 0.481
NRMSE 0.677
NRMSE 0.103
NRMSE 0.357
NRMSE 0.15
NRMSE 0.177
NRMSE 0.796
NRMSE 1.52e-16
Cantera
NN mean of 3
failed 1.058 ms
NRMSE 0.043
NRMSE 0.109
NRMSE 0.339
NRMSE 0.604
NRMSE 0.203
NRMSE 0.434
NRMSE 0.0378
NRMSE 0.766
NRMSE 0.664
NRMSE 0.00313
Time (ms)

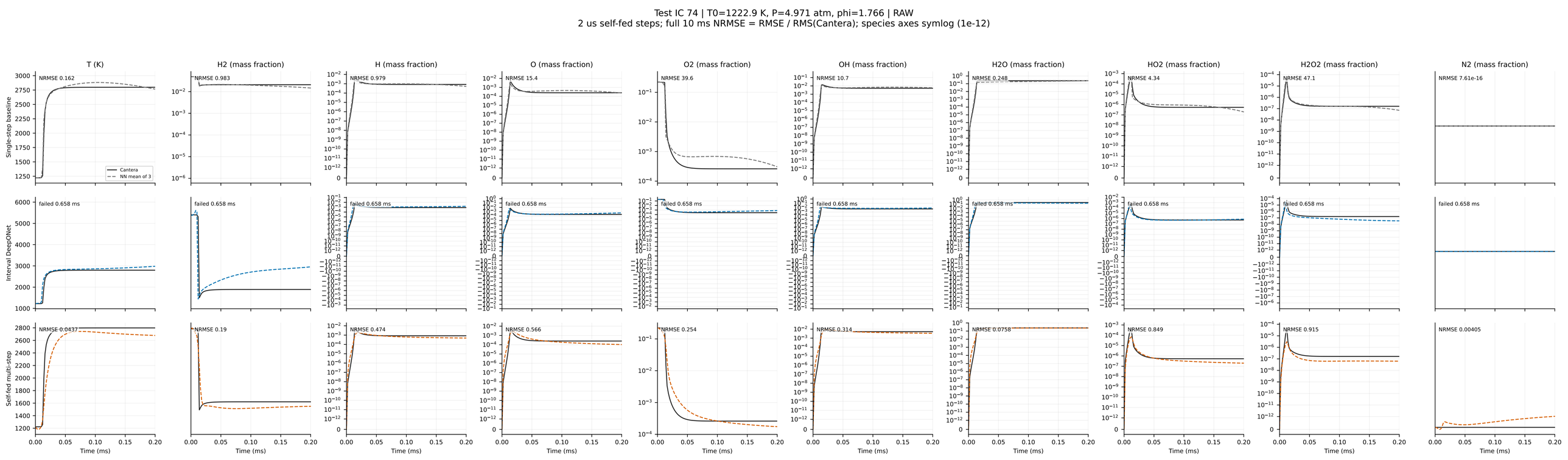

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 1.664 ms
failed 1.634 ms
Cantera
NN mean of 3
NRMSE 0.0408
NRMSE 1.94
NRMSE 1.8
NRMSE 0.214
NRMSE 0.561
NRMSE 0.22
NRMSE 0.295
NRMSE 0.328
NRMSE 0.131
NRMSE 0.0065
Time (ms)

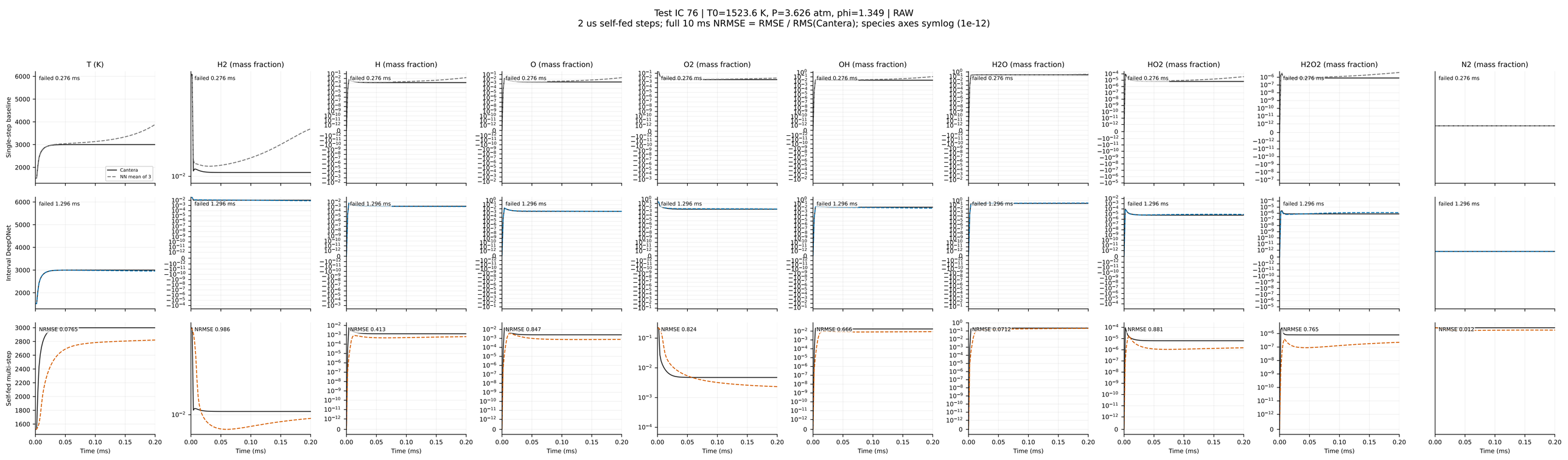

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.517
NRMSE 0.94
NRMSE 8.93
NRMSE 5.49
NRMSE 0.535
NRMSE 2.71
NRMSE 0.644
NRMSE 0.88
NRMSE 0.996
NRMSE 1.47e-16
failed 0.02 ms
NRMSE 0.455
NRMSE 0.961
NRMSE 11.3
NRMSE 2.9
NRMSE 0.62
NRMSE 2.06
NRMSE 0.796
NRMSE 0.89
NRMSE 0.997
NRMSE 0.00646
Time (ms)

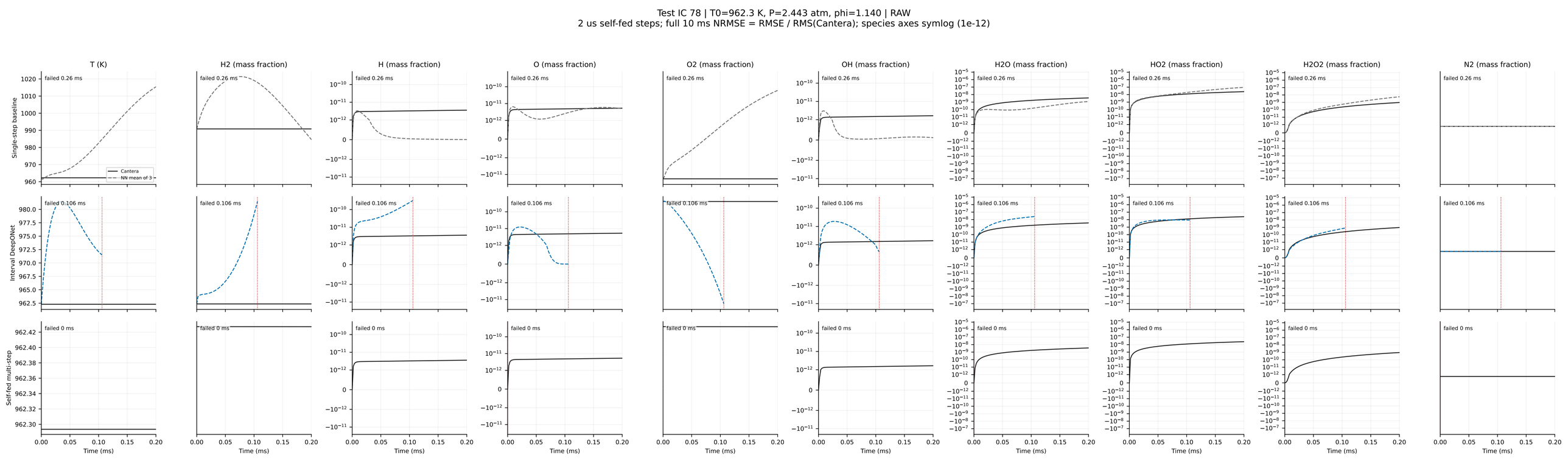

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.528 ms
failed 0.116 ms
failed 0.722 ms
Cantera
NN mean of 3
Time (ms)

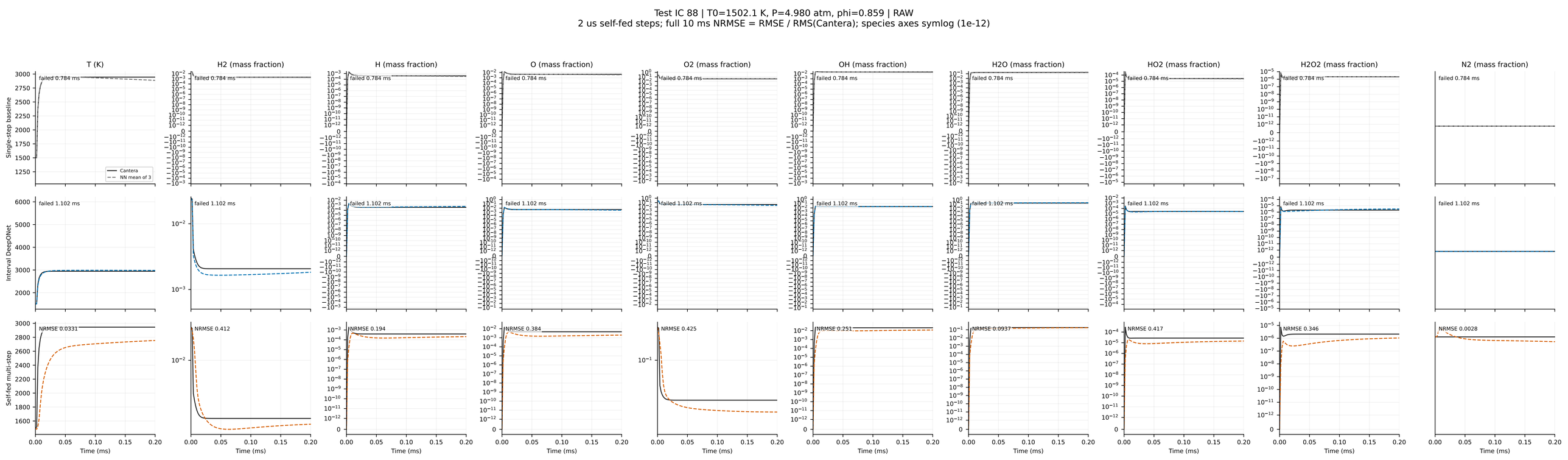

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.02 ms
failed 0 ms
Cantera
NN mean of 3
+9.3496e2
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0487
NRMSE 0.131
NRMSE 0.381
NRMSE 0.963
NRMSE 0.214
NRMSE 0.532
NRMSE 0.0754
NRMSE 0.534
NRMSE 0.638
NRMSE 1.53e-16
failed 0.32 ms
NRMSE 0.012
NRMSE 0.0237
NRMSE 0.163
NRMSE 0.43
NRMSE 0.112
NRMSE 0.109
NRMSE 0.0398
NRMSE 0.425
NRMSE 0.518
NRMSE 0.000697
Cantera
NN mean of 3
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.162
NRMSE 0.983
NRMSE 0.979
NRMSE 15.4
NRMSE 39.6
NRMSE 10.7
NRMSE 0.248
NRMSE 4.34
NRMSE 47.1
NRMSE 7.61e-16
failed 0.658 ms
NRMSE 0.0437
NRMSE 0.19
NRMSE 0.474
NRMSE 0.566
NRMSE 0.254
NRMSE 0.314
NRMSE 0.0758
NRMSE 0.849
NRMSE 0.915
NRMSE 0.00405
Cantera
NN mean of 3
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
failed 0.276 ms
failed 1.296 ms
Cantera
NN mean of 3
NRMSE 0.0765
NRMSE 0.986
NRMSE 0.413
NRMSE 0.847
NRMSE 0.824
NRMSE 0.666
NRMSE 0.0712
NRMSE 0.881
NRMSE 0.765
NRMSE 0.012
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.26 ms
failed 0.106 ms
failed 0 ms
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | RAW
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
failed 0.784 ms
failed 1.102 ms
NRMSE 0.0331
NRMSE 0.412
NRMSE 0.194
NRMSE 0.384
NRMSE 0.425
NRMSE 0.251
NRMSE 0.0937
NRMSE 0.417
NRMSE 0.346
NRMSE 0.0028
Time (ms)